

Making Logical Inductors satisfy the Reflection principle and Epistemic self-trust

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Reflection principle: Beliefs about future beliefs marginalize to present beliefs.

Epistemic self-trust: Conditioning on a future belief that the probability of φ is p , the probability of φ turns p .

We cannot just ban some worlds from the betting table and be done with it (as with the simpler coherence properties), since for example, given φ is true, it's perfectly coherent to have $\mathbb{Q}_{k+n}(\varphi) = 0.4$, and also perfectly coherent to have $\mathbb{Q}_{k+n} = 0.8$ (even if, if we trust $\mathbb{Q}$'s epistemic evolution, the latter should a priori seem more likely). And nonetheless, if the price of $\varphi \wedge \mathbb{Q}_{k+n}(\varphi) = 0.4$ were too high, Reflection would break. So this is a global property of $\mathbb{Q}_k$, instead of a local property of single possible worlds that we can remove from the betting table.

To obtain these properties, we'll only let traders bet on the meta-beliefs (beliefs about future beliefs) up to a finite future horizon, and from them (by marginalization) extract their object-level beliefs about basic logic. That is, the Reflection principle is only sometimes broken because traders can bet independently on object-level beliefs (φ) and on future beliefs about them ($\mathbb{Q}_{k+n}(\varphi) = p$). If we only let them bet on the latter, and from those deduce by marginalization the object-level prices of sentences, we'll obtain marginalization by default. And we can still, as above, let traders bet on object-level sentences: betting on φ will just amount to betting on many worlds where $\mathbb{Q}_{k+n}$ is high.

$n = 1$ case:

To see this more clearly, consider the case where $n = 1$. Say our traders can only bet on (possible worlds which are conjunctions of possibly negated) sentences of the form " $\mathbb{Q}_{k+1}(\varphi) \in (\frac{m}{2^{k+1}}, \frac{m+1}{2^{k+1}}]$ " (that is, instead of allowing arbitrary precision of the form " $= p$ " for any rational p , we use a coarse-graining, that gets more precise with time). Then, when a trader wants to buy an object-level share of φ , we actually sell it the bundle with $\frac{m+0.5}{2^{k+1}}$ shares of " $\mathbb{Q}_{k+1}(\varphi) \in (\frac{m}{2^{k+1}}, \frac{m+1}{2^{k+1}}]$ ". That is, we distribute its wealth so that future worlds in which φ seems more probable are more likely. And finally, once all traders have placed their bets and the market prices are set, we can just define

$$\mathbb{Q}_k(\varphi) := \sum \frac{m+0.5}{2^{k+1}} \mathbb{Q}_k(\mathbb{Q}_{k+1}(\varphi) \in (\frac{m}{2^{k+1}}, \frac{m+1}{2^{k+1}}])$$

That is, *Reflection principle* by definition. And then, when there's a mixture of object-level and

meta-level sentences, by definition, we also set

$$\mathbb{Q}_k(\varphi \wedge \mathbb{Q}_{k+1}(\varphi) \in (\frac{m}{2^{k+1}}, \frac{m+1}{2^{k+1}}]) := \frac{m+0.5}{2^{k+1}} \cdot \mathbb{Q}_k(\mathbb{Q}_{k+1}(\varphi) \in (\frac{m}{2^{k+1}}, \frac{m+1}{2^{k+1}}])$$

That is, *Epistemic self-trust* by definition.

And then, the other simpler coherence properties will be implemented in the object-level beliefs of $\mathbb{Q}_{k+1}$: any possible world we are considering (in which $\mathbb{Q}_{k+1}$ turns out to have this set of opinions) is one where $\mathbb{Q}_{k+1}$ is Coherent (that is, we eliminate those which are not from the betting table). And as a consequence, through marginalization, those Coherence properties will also be translated into our derived object-level beliefs.

General n case:

I think we can do a similar setup for the general n case, but I haven't formalized it yet. The additional complication is that now the meta-beliefs of future different stages (for example, $\mathbb{Q}_{k+1}$ and $\mathbb{Q}_{k+2}$) also need to marginalize between themselves.

The idea is to consider only "bundles" of probabilities over future trajectories that correctly marginalize (that is, we take the "incoherent" trajectories, in the sense that they don't correctly marginalize, off the betting table). And we only let traders buy these bundles.

Important note: Traders are no longer trading on possible worlds. A possible world here is a concrete trajectory of the future beliefs. Now, traders are buying and selling pre-made "bundles", which are probabilities over possible worlds. We, as bundle dealers, ensure that all these bundles ensure the required properties, and so the aggregate economic movement that a trader performs (a weighted sum of bundles) also satisfies them (I need to prove that the set of bundles is convex).

And as above, buying a share on an object-level belief will amount to buying shares on bundles according as to how much they ratify that belief.